Proposed Title and Core Message **Questioning Artificial Intelligence**: How the Art of Asking Questions Defines AI and Helps Navigate an Uncertain Future

Brief Blurb When we ask either or a student or an AI questions, we learn how smart they are. This book helps connect the arcana of how to build AI systems and extract answers from Wikipedia to an insider's view of high-stakes trivia games.

By drawing those *connections* between the two worlds of human and computer question answering, this book shows how human expectations of QA have shaped AI, how there's still more to learn from human QA, and how a more mature AI landscape may change how humans answer questions for fun and in the jobs of the 21st century.

Keywords

- Question Answering
- Artificial Intelligence
- Item Response Theory
- Natural Language Processing
- Trivia
- Standardized Tests
- Leaderboards

Author Information I'm excited to be writing this because it's something that I'm passionate about: answering questions is central not just to my research program but also to my day job as a professor at the University of Maryland: teaching and guiding students. And it's also important to me as a hobby: even before this became part of my research program, I spent weekends writing and answering silly trivia questions. I'm hoping that this will help more people appreciate the joy and beauty of posing and answering questions.

In my day job, I have built question answering systems that used the structure of Wikipedia to better answer questions, was one of the first to apply deep learning to the QA task, and played a lot of exhibition games against trivia whizzes like Yogesh Raut, Ken Jennings, and Roger Craig. I'm also a

professor of computer science, and one of the courses that I teach is a course on question answering (and this book is the textbook for it).

As a hobby, I'm also a trivia enthusiast. I played QB for Caltech (where I got BS degrees in computer science and history), started Princeton's pub quiz (where I went to grad school), and appeared on *Jeopardy!*. To be clear, I'm not very good at trivia: our QB teams at best cracked top ten nationally, and I came in second place in the one game of *Jeopardy!* I taped (my students are particularly fond of mocking my poor performance on the video game category). But this has brought me a familiarity with the trivia community and how it works, one that I think gives me a unique (or deranged) look at how computers answer questions.

In contrast, I've won more awards as a researcher. Probably the one I'm most proud of is the Karen Spärk Jones award (we'll learn more about her in Chapter 4) from the British Computing Society, but I've also gotten an NSF CAREER award and "best of" awards from Intelligent User Interfaces, Neural Information Processing Systems, the North American Association for Computational Linguistics, Empirical Methods in Natural Language Processing, and the Conference on Natural Language Learning. I've published over a hundred peer-reviewed publications on interactive machine learning, question answering, and exploring document collections; and this isn't my first book, with David Mimno and Yuening Hu, I wrote *Applications of Topic Models*. Although I should say that because I'm a professor, this is mostly the work of the students that I've worked with over the years (I thank them by name at the end of the book).

But the biggest prize that I've gotten is to have a career (and permissive bosses) that allow me to combine my hobby with these intellectual pursuits. And I'm even more lucky that that career allows me to now share it with you.

Text Type and Market The book is intended as an "armchair scientist" in the general public to learn about human and computer question answering. or for machine learning researchers to learn how *human* question answering can improve their computer question answering systems.

For example, a trivia enthusiast curious about how their hobby relates to modern AI could benefit from the book and see how the style of asking questions perfected by years of iterative refinement in quiz bowl tournaments, game shows, and pub quizzes show that maybe AI isn't as good as it says it is.

Likewise, a more casual reader who is familiar with trivia games (but not necessarily an expert) might see these accessible, human-centered games as a scaffold to better understand the challenge of knowing whether a new AI is good or not. It will also show the development of AI over the last few decades compared to human abilities.

Moreover, the book can serve as a "unifying" textbook in courses. Although it is not technically deep, it could be used in:

- A freshman seminar introducing students to AI through the lens of question

answering.

- A graduate seminar in information seeking or evaluation in information science.
- A companion to technical papers in a graduate computer science course. And this is how I have used chapter drafts: as an overview to show the "big picture" of the technical papers we read.

The book is based on a course that I teach at the University of Maryland. The lectures are available online, and the most common international visitors are from:

- India
- China
- Germany

Competition

- Ken Jennings. *Braniac: Adventures in the Curious, Competitive, Compulsive World of Trivia Buffs*. Villard, 2007.
- Mark T. Maybury. *New Directions in Question Answering*. AAAI Press, 2004.
- Navin Sabharwal and Amit Agrawal. *Hands-on Question Answering Systems with BERT*. A Press, 2021.

These books are different because they either focus on *human* question answering or *computer* question answering, but not drawing the connection between the two.

Length and Schedule Estimated book contents:

- 150 Single-spaced pages
- Two dozen diagrams / illustrations
- Draft to reviewers July 31, 2026, revisions/edits until December 31, 2026

Three chapters were published as position papers. If accepted, the chapters will be substantially revised to flow well with and complement other chapters.

Sample chapter to follow.

How the Book is Structured While this book is obstensibly about how *computers* answer questions, that is not where the book begins. This history is important because it explains why humans value answering questions. This provides historical context on why humans equate answering questions correctly with

intelligence but also because key elements for artificial intelligence—item response theory and preference models—were developed from looking at how humans answered questions.

The next stop is looking at *today's* question answering, begining with Alan Turing, arguably the father of computing and artificial intelligence. Two wartime efforts—one at air and one at sea—began a rivalry in question answering, which we call the Cranfield and Manchester based on the college towns home to Alan Turing's theorizing on intelligent computing and Cyril Cleverdon's practical evaluation of computing systems. Having set the stage of this rivalry, we then see how these ideas built the foundations of modern AI from the methods that let *Watson* win on *Jeopardy!* (this is the subject of the sample chapter for the proposal) to datasets like Google's Natural Questions that set the stage for smart chatbots and the models like GPT.

Finally, the book closes with the *future* of question answering, focusing on what it means for an AI to be intelligent: how to turn Alan Turing's thought experiment into reality and how to make sure that computers don't trick us into thinking they're intelligent (but just cheating on the test). And if computers are becoming intelligent the question is how to *align* them with users, which sees the return of item response theory and preference models: this time built up so we can better understand why AI says the things it does. Finally, we close with a discussion of the "big questions" that computers cannot answer on their own: how can the models developed from answering questions from humans best work alongside humans for work and play.

1
Chapter Synopses

A brief description of each of the proposed chapters.

1: Introduction: The Unanswered Questions

Since this is a book about question answering, I should try to answer some questions to start things out. I'm going to try to answer: "what's this book about", "who am I", and "who can use this book".

2: Question Answering: The Foundation and Savior of Civilization

Despite its central importance to AI, asking and answering questions is not new—it stretches back into our collective unconscious. Riddles and trivia have persisted for millennia not just because they're entertaining. Indeed, being able to ask and answer questions is close to godliness. From the Sphynx to Gestumblindi, this chapter examines how myths from several cultures connect answering questions to forming identity, unlocking the secrets of the universe, and gaining intelligence. And connecting this to nascent "superhuman" AI. Later, beyond proving worthiness in myth, asking questions helped combat corruption and nepotism, the downfall of multiple civilizations. However, the administration of carefully formed questions saved China and the United States via the reforms from Wu Zetian to the Pendelton Act. Exams created meritocracy and social mobility while also building an understanding of what it means to write a "fair" and "useful" question.

Topics in chapter:

1. **The Riddle of the Sphynx: Questions that Define Destiny.** Introduces question answering through myth: the Sphynx's riddle as an early example of how questions can determine life, death, and human identity.
2. **Socratic Questions as a Teaching Tool.** Explores the Socratic method as a foundational model of question answering, where dialogue and probing questions serve as a tool for reasoning, learning, and uncovering hidden assumptions.
3. **Only a God Would Ask Questions Like that, "Gestumblindi".** Examines the Old Norse riddle contest of Gestumblindi, highlighting how question answering has long been tied to wit, deception, and the boundary between human and divine knowledge.
4. **How Exams Saved China and the US.** Shows how large-scale examination systems in imperial China and later the United States institutionalized question answering as a mechanism for governance, meritocracy, and social power.

3: The Turing Test: The Deceptions that Defined Artificial Intelligence

In the 20th century, we turned the instinct of measuring worth through questions toward machines. In the 1950s, Alan Turing proposed a parlor game that would come to define artificial intelligence: could a wily interrogator discern whom they were talking to just through posing clever questions. The eponymous Turing Test is the most durable (but contentious) definition of what an intelligent computer is, but it is built on deception.

Topics in chapter:

1. **Turing's Legacy.** Why Turing is widely considered the father of computer science for multiple reasons and how he contributed to cloak and dagger spy operations.
2. **The Imitation Game.** A gay man in England—where homosexuality is criminalized—posits a game about the ability of someone to pass in different gender roles. Why this is difficult lays the foundation for why computers still struggle to capture human nuance.
3. **No, the Turing Test has not been Solved.** While many have claimed to pass the Turing Test, it hasn't happened yet. We review some notable claims of passing the Turing Test and why they fell short, and why you shouldn't fall for the claims of charlatans.
4. **A Rigorous Test.** The ingredients that make a Turing Test legitimate
5. **General Artificial Intelligence.** How the holy grail of AI is defined and how the Turing Test can measure it.

4: The Cranfield Paradigm: How a University with an Airstrip made Google Possible

Modern search engines feel inevitable, but they emerged from a radical mid-century idea: that question answering systems could be evaluated scientifically using reusable test collections. While the Turing test is a *conceptual* measurement, this chapter traces the origins of the Cranfield paradigm's *empirical* measurement of machines' ability to answer questions. It introduces precision and recall as the core metrics of retrieval, and shows how sparse term-based methods like tf-idf laid the groundwork for large-scale evaluation efforts at NIST and ultimately for web search as we know it. It also situates these retrieval advances within the broader trajectory of "good old fashioned" AI approaches to question answering.

Topics in chapter:

1. **Cleverdon's Clever Collection.** Introduces Cyril Cleverdon's Cranfield experiments, which replaced librarian-crafted indexing with a measurable framework for evaluating retrieval systems using shared document collections and relevance judgments.
2. **Is it Better to Find All the Good Things or for Everything Found to be Good?.** Presents the classic contingency-table view of retrieval evaluation, defining precision and recall as the fundamental tradeoff between finding relevant documents and avoiding irrelevant ones.
3. **Which Words Matter Most in your Index?.** Explains how term-frequency and inverse document frequency weighting, developed by Karen Spärck Jones, transformed Cranfield's evaluation insights into practical retrieval methods that could scale beyond curated catalogs.
4. **Old-Fashioned AI's Search for Meaning.** Connects information retrieval to symbolic AI traditions, where question answering was framed as logical inference and knowledge representation rather than ranking documents by statistical similarity. Discusses Green et al.'s 1979 Baseball system; Woods and Kaplan's 1972 LUNAR system; and Winograd's SHRDLU.
5. **Making Hide and Seek (in Documents) a Professional Game.** Describes how NIST and the TREC competitions institutionalized the Cranfield paradigm, building larger test collections that enabled systematic progress and shaped the evaluation culture behind Google-era search.

5: The Manchester Paradigm: The Art of Asking the Perfect Question

While computer scientists were getting their feet wet answering questions in the wake of World War II, trivia enthusiasts were perfecting how to ask the perfect question. This chapter outlines the conventions and processes of the highest form of question answering—in the biased opinion of the author—and argues for why many of these standards could be a part of creating data for computer question answering. Previously published as Boyd-Graber and Börschinger (2020).

Topics in chapter:

1. **Two British University Towns Alike in Question Answering.** In contrast to the existing Cranfield Paradigm, we propose the Manchester paradigm, which creates probing questions to test specific skills of AI systems.
2. **It Takes a Village to Ask a Question.** I review the Quizbowl—whose University Challenge incarnation is one of the reasons we adopt the name Manchester paradigm—and how it reflects the incremental perfection of question answering for humans and what lessons (e.g., adjudication, focusing on discriminative questions) can transfer to computer question answering.
3. **The Craft of Question Writing.** A good trivia question should be unambiguous, discriminative, and test knowledge. We review how these norms developed and are executed in crafting questions for humans.
4. **Ignore the Distinction at Your Peril.** Why the distinction between Cranfield and Manchester is relevant for understanding AI ability.
5. **Your Trivia Tournament is a Leaderboard.** Leaderboards in question answering are not fundamentally new: they are accuracy statistics that induce rankings, much like trivia tournaments. This section argues that QA evaluation can learn from decades of human competition norms about fairness, clarity, and discrimination.

6: Watson on Jeopardy!:
Unquestioned Answers from IBM's tour de force

It's been nearly a decade since IBM Watson crushed two puny humans on Jeopardy! Some people took that to mean that computers were definitively better than humans at trivia. But that isn't the complete answer—this chapter, inspired by Jeopardy!'s gimmick of responding to answers to questions, questions some of the "answers" that emerged from IBM's tour de force.

Topics in chapter:

1. **Why IBM Chose Jeopardy! for a Grand Challenge.** Places Watson in the tradition of AI grand challenges such as Deep Blue and AlphaGo, emphasizing how these projects are designed to create cultural as well as technical shifts in the understanding of machine intelligence.
2. **How Watson Works.** Introduces Watson as a statistical QA system built from retrieval, feature engineering, and probabilistic decision-making rather than modern neural generation.
3. **This Game Is Rigged, I Tell Ya!.** Presents the chapter's central critique: Watson's victory was not a clean scientific comparison, because the match conditions systematically advantaged the machine.
4. **Two Nice Guys, One Computer with no Shame.** Watson is not just a system that provides answers to questions. It also needs to decide which questions to answer and whether to trust a guess. How does it decide these, and how does that contribute to Watson's perceived human domination.
5. **The Legacy of Watson.** Concludes that Watson marked both a milestone in question answering and a cautionary tale about evaluation, motivating the need for fairer benchmarks and clearer measures of progress.

7: Formats Beyond Stand-Alone QA

How do you ask a question: is it text, a picture, or a conversation? This chapter reviews the different forms question answering can take and what complexities that can introduce.

Topics in chapter:

1. **Task vs. Format.** Introduces the distinction between what question answering systems are trying to accomplish (the task) and the external shape that QA takes in practice (the format). The chapter argues that many of the hardest problems in QA emerge not from new tasks, but from new formats that demand context, interaction, or multiple kinds of evidence.
2. **What is a Question?.** Explores how questions vary far beyond short factoid prompts, ranging from multihop queries to conversational follow-ups and long-form explanations. The section emphasizes that defining what counts as a "question" is itself a modeling challenge, because real information needs are often underspecified, contextual, and evolving.
3. **Where do you get an Answer?.** Discusses the role of evidence in QA systems, contrasting settings where answers are directly extractable from a single passage with formats that require gathering multiple supporting documents. The section frames retrieval, linking, and reasoning over evidence as central bottlenecks for moving beyond one-hop machine reading.
4. **What did you Mean?.** Focuses on conversational question answering, where users ask sequences of context-dependent questions that require memory, clarification, and ambiguity resolution. The section surveys datasets like QuAC, CoQA, and CANARD, and highlights the evaluation difficulty: conversational success depends on interaction quality, not just matching a reference answer.
5. **A Picture is worth a Thousand Questions.** Extends QA beyond text to multimodal settings, where systems must answer questions about images, video, or audio. The section introduces visual QA datasets such as VQA and VizWiz, explains how transformers incorporate visual objects as inputs, and argues that evaluation becomes harder when answers are not naturally grounded in text spans.
6. **Questions Beyond English.** Briefly motivates the need for QA systems that work across languages and cultural contexts, where named entities, scripts, and question conventions differ substantially. The section notes that multilingual QA raises both technical challenges in retrieval and deeper questions about what constitutes an acceptable answer across linguistic communities.
7. **The Skills Needed to Answer Questions.** Synthesizes the diverse capabilities required for robust QA: retrieval, coreference resolution, multistep reasoning, and dialog management. The section emphasizes that QA is not a single problem but a bundle of interacting skills.
8. **A Taxonomy of Question Answering.** Concludes by organizing QA formats into a broader taxonomy, showing how multihop, conversational, long-form, multimodal, and spoken QA each stress different assumptions about questions, evidence, and evaluation. The section argues that progress in QA depends on matching models and metrics to the true complexity of human information seeking.

8: Datasets: Build or Adopt?

The cliché is that data are the new oil, powering AI. Fortunately, because humans naturally ask questions, there are many datasets that we can find 'for free'. However, these datasets still come at a cost: many of these datasets have inherent problems (e.g., ambiguities and false presuppositions) or oddities (only talking about American men) that make them difficult to use for question answering. This chapter discusses these datasets that have formed the foundation of much of modern AI. If you can't find the data you need, build it yourself. But this is not always a perfect solution. This chapter discusses the datasets that people have built and the new problems that this can create.

Topics in chapter:

1. **Constructed Datasets from Crowdsourcing.** The SQuAD dataset helped create the first wave of question answering. It also paved the way for other constructed datasets built from workers on the web.
2. **Natural Questions Five Years Later.** Google's Natural Questions dataset represents the most realistic dataset to question answering, but is it a good way to train computers?
3. **Turning to Experts.** Another option is to repurpose datasets created by experts designed to test humans.

9: The Tools to Answer Questions

Having discussed how we create datasets that can teach computers how to answer questions, we now explore how those datasets can train modern computer methods.

Topics in chapter:

1. **Modern Knowledge.** This section reviews how we can turn natural language queries into actionable queries in databases.
2. **Johnny 5 Can't Read.** However, putting information in a database is difficult and time-consuming…not all information is in a database (indeed, some information defies strict database schemas). Thus, we need to teach computers how to read. This chapter reviews the process of "machine

reading", where computers find information in a large text corpus and then extracts the answer from it.

3. **A Vector in a Haystack.** As deep learning became practical, the field has moved from representing text with discrete words to continuous vectors. This is also the case for question answering. This chapter reviews how these representations can help us find relevant answers to questions.
4. **We Couldn't Find an Answer, so we had to Generate One.** The deep learning revolution not just helps us find answers but to generate them. This chapter talks about the promise and peril of these approaches: how we can synthesize much richer information, make stuff up, encourage these agents to align to our wishes using RLHF, and how it's hard to tell if an answer is any good.

10: Siri takes the SAT

How do we know how smart a machine is? Like a human, we typically give it a test; the difference is that tests for computers are called 'leaderboards'. This chapter talks about the pros and cons of leaderboards and how some of the methods used to analyze human standardized tests can help us understand the strengths and weaknesses of not just computer question answering specifically but AI generally. This also marks the reappearance of item response theory, which can help make deciding the smartest computer more efficient. Previously published as Rodriguez et al. (2021) and **?**.

Topics in chapter:

1. **When 21 questions are worthless.** We compare how a trivia cheating scandal extends to AI leaderboards. Questions have been memorized, the question askers offer tells and signals so that a Clever Hans can still answer the question without understanding, and exams don't necessarily measure what they claim to.
2. **The Hype and Confusion of Leaderboards.** When a new model comes out, there is usually a hullaballoo about how it's the 'best ever'. But how do you know? This is a job for a leaderboard, but these aren't always up to the task.
3. **The Balance Between Skill and Difficulty.** In the 1920s, Ernst Zermelo came up with a formula to determine, given a chess tournament who was the best player. This formula, building on logistic regression, became the foundation for understanding not just Elo rankings but political ideology and students' skill when taking the SAT. It also gave birth the Bradley-Terry models needed for RLHF.
4. **Is this Question Bad or am I just Stupid?.** A leaderboard is like a game show. If you keep every player's score, it can give you clues not just about how good they are but whether questions are good or not. And once you

know what the best questions are, you can deploy them more judiciously.

5. **Sure, you can add fifteen digit numbers, but can you identify a extitSimpsons plot from a vague description?.** If we think about skill as a vector (i.e., I can be good at math but bad at chemistry), then we can have a more nuanced comparison between those who answer questions. This allows for a clearer comparison between computer agents but also between humans and computers: computers are good at raw fact retrieval, but humans are better at matching abstract patterns and making inductive "leaps".
6. **Easy for Humans, Hard for Computers.** If existing datasets and game show appearances aren't enough to tell whether humans or computers are better at answering questions, what can we do? While there is an argument for focusing on natural data, modern language models are changing not just what is possible computationally but changing the language itself. Thus, we need to select examples specifically to challenge computers. These examples are called *adversarial examples*, and this chapter presents how to gather them and how they can reveal the strengths and weaknesses of computer question answering.
7. **Augmenting Humans, not Replacing Them.** But perhaps we shouldn't be measuring humans and computers on the same yardstick to see which is better. A more realistic measurement that better measures how humans will be working with AI is seeing how well humans and computers can work together to answer a question.

11: Finding and Dealing with Bad Questions

This chapter discusses how questions can be imperfect and how both humans and computers deal with those imperfect questions. Navigating an imperfect question requires more skill than answering a perfectly posed one. Not only must the answerer have the requisite information about the correct answer, they must exclude incorrect interpretations and navigate how to decide what the asker actually needs.

Topics in chapter:

1. **What's the Capital of Georgia?.** When you hear a question, there can be multiple interpretations. How to resolve those interpretations often requires as much mental agility as actually answering the question.
2. **When did you stop beating your wife?.** Sometimes questions assume something that they shouldn't. Detecting these "false presuppositions" is tricky enough for humans, but modern AI is particularly gullible and often plays along when it shouldn't.
3. **What city is the setting of Crime and Punishment?.** Even if a question is well-formed, sometimes the problem lies with the candidate answers or

detecting when an answer is correct. This is a problem that seems easy at first, but the devil is in the details.

4. **What do you do with a broken question?.** One of the challenges humans are able to overcome with relative ease is navigating broken questions to get to the answer that the asker actually wants. We discuss what strategies humans use and how computers can learn from them.

12: Fulfiling Turing's Gameshow

The year is 2025, and there's a new gameshow that not only showcases the most advanced AI available but also keeps the public informed about the limitations and struggles of current technology. This chapter outlines the ten seasons of the show and how it tracks the development of machine intelligence, leading to passing the Turing Test.

Topics in chapter:

1. **Where there's room for improvement.** One of the core arguments of the book is that AI can learn from trivia and game shows. What if we put our money where our mouth is and followed through? How could we improve AI by taking the best of both worlds and made it entertaining?
2. **Moving beyond White American Men.** Many datasets—both Cranfield and Manchester—overwhelmingly talk about white American men. Is that a problem, and if so, how might we improve dataset representation and AI's understanding of other people and places.
3. **Some Rules aren't Written.** One of the reasons I'm still a fan of the Turing test is that it put implicit social norms as a key part of the test. This has not been incorporated into major AI evaluations, but it should.
4. **Making the Turing Test Real (and Fun).** We conclude the chapter with a roadmap for how an AI game show could grow and evolve into a replicable, entertaining, and effective test of AI intellectual and social intelligence that Alan Turing would recognize as a descendant of his thought experiment.

13: What AI Dystopias You Should be Afraid of

While the book began with how question answering in myth helped the ancients grapple with a changing and uncertain world, today's fictions represent our contemporary attempts to understand the advent of intelligent computers. This chapter reviews human–computer question answering in 2001, Star Trek, The Terminator, Blade Runner, Futurama, and what these depictions reveal about both human conceptions of artificial intelligence and how it might shape future deployments of question answering.

Topics in chapter:

1. **Computer, speculate.** A review of the role of QA in science fiction: Voigt-Kampf, Star Trek, and Futurama have all used humans questioning computers to test how human or intelligent they are. Do these portrayals offer lessons for the real world?
2. **p(Doom).** Beyond the imagining of science fiction, what are the actual downsides to the widespread availability and deployment of 'intelligent' AI? This chapter talks about AI's ability to amplify existing negative *human* tendencies in responding to questions and how—while dangerous and worthy of mitigation—don't demand us to halt the development of modern AI.
3. **Fear of a Bot Planet.** I conclude the book with a discussion of what the risks of widespread AI deployment actually are and what questions we should be asking of people who are advocating for increasing the role of AI.

Chapter 6

Watson on Jeopardy!: Unquestioned Answers from IBM's tour de force (Sample Chapter)

It's been nearly a decade since IBM Watson crushed two puny humans on Jeopardy! Some people took that to mean that computers were definitively better than humans at trivia. But that isn't the complete answer—this chapter, inspired by Jeopardy!'s gimmick of responding to answers to questions, questions some of the "answers" that emerged from IBM's tour de force.

Topics in chapter:

1. **Why IBM Chose Jeopardy! for a Grand Challenge.** Places Watson in the tradition of AI grand challenges such as Deep Blue and AlphaGo, emphasizing how these projects are designed to create cultural as well as technical shifts in the understanding of machine intelligence.
2. **How Watson Works.** Introduces Watson as a statistical QA system built from retrieval, feature engineering, and probabilistic decision-making rather than modern neural generation.
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5. **The Legacy of Watson.** Concludes that Watson marked both a milestone in question answering and a cautionary tale about evaluation, motivating the need for fairer benchmarks and clearer measures of progress.

We begin the "modern" age of computer question answering and the rise

of AI with *Watson*: an IBM-built AI that could play the American game show *Jeopardy!*

It's been over a decade since *Watson* appeared on TV, but the passage of time has not dulled the luster of this revolutionary *tour de force*. This chapter will talk about how it changed my life, how it changed the way the public thought about AI, and how its technology paved the way for "modern" AI as exemplified by LLMs like GPT.

But this isn't a love letter to *Watson*. For all the fame and glory it won, the *Watson* story isn't without problems. Indeed, this chapter will make the argument that the game was subtly rigged in a way that set back research on AI.

6.6 Why IBM Chose *Jeopardy!* for a Grand Challenge

Watson is part of a long history of "grand challenge" projects meant to symbolize progress on AI. IBM (International Business Machines) is a New York-based technology company that for much of the twentieth century was synonymous with computing.

It had previously wowed the world by creating "Deep Blue", a chess-playing computer that defeated Gary Kasparov in 1997, widely considered the best chess player of the time (Hsu, 2002). Other AI grand challenges include those issued by funding agencies like the United States' Defense Advance Projecects Research Administration (DARPA) to build autonomous cars that can drive a hundred miles in the Mojave, although the best team just managed over seven miles (Patterson, 2005). Google/DeepMind had a similar ambition when they created AlphaGo, which like DeepBlue defeated Lee Sedol in 2016, widely considered the best Go player of the time (Koch, 2016).

The goal of these projects is not just to advance technology but to create a big enouch splash that people change how they think about technology.

It's hard to imagine an alternate world where these grand challenges did not happen, but I don't think it's exaggeration to say that these grand challenges did indeed change the world. After his defeat, Gary Kasparov took a "if you can't beat them, join them" approach and then began arguing for Centaur Chess: humans and computers playing chess together (this works for question answering too, as we discuss in Chapter 10.7). Likewise, the best Go players became more creative and novel moves by incorporating computer-like play into their play style **?**. And two decades after DARPA's autonomous grand challenge, we now have self-driving cars on the street in several US cities.

So what made *Jeopardy!* the right pick for this grand challenge?

What Jeopardy! is and How it Works

Jeopardy! is a gameshow created by Merv Griffin that debuted in 1967 (Griffin, 2003). Its big gimmick is that the player responses are given in the form of a

question in reaction to infamous human cheating scandals (we discuss how computers "cheat" in Chapter 10). For example, the clue

> The CAPTCHA test against spam & robot programs is called the 'reverse' test named for this British code breaker

would have response Who is Alan Turing. The clues are arranged in a grid: columns represent categories and rows represent difficulty, with the more difficult questions being worth more.

There are three players who stand side by side behind podiums. When a clue is read, any of the three players can "buzz in" to say that they want to give a response. If they give the correct response, they then have "control of the board" and can select the next clue.

One advantage of controlling the board is that some questions are called "Daily Doubles" which allow the player to potentially double their score: a player can wager any part of their score, and if they get it right they get that ammount added to their score (but a wrong response will subtract it from their score).

How Watson changed my life

I remember when I first heard rumblings of Watson. Because I had a foot in both the AI and trivia communities, I heard two different stories. I heard rumors of amazing work in parsing and semantic role labeling happening from researchers who ventured north to Westchester county in New York (I was doing my PhD in New Jersey). From the trivia community, I heard of some people who were being paid by IBM to play trivia games but that they couldn't say anything more.

I was very sceptical. By the time that *Watson* came to fruition, I had moved to the University of Maryland. Then, my scepticism turned to jealousy. I watched, along with the rest of the world, one of the greatest achievements of AI unfold in front of me. What was I doing wasting my time working on language models (?) if this was also legitimate research?

Let me be clear that the technical triumphs are indisputable (and, in my opinion, under-appreciated). From the work on wagering to synthesizing multiple information sources, Watson (Ferrucci et al., 2010) was from top to bottom a top-notch well-oiled machine. And it computed all of this in real time—something that wasn't strictly necessary but still impressive.

6.7 How Watson Works

While "neural" question answering (which we'll discuss in future chapter) is the big thing these days, Watson came of age in the *statistical* age. To understand some of how Watson works, its helpful to review some of the work that came before Watson that helped inspire it.

Rule-based Question Answering

Preceeding the statistical age of QA was the *rule-based* age: systems that were impressive on individual questions about baseball or geology but that faltered as soon as it saw a question that was unexpected in terms of the domain, they'd fail miserably.

One of the guiding principles of the statistical age was "the unreasonable effectiveness of big data", and this required scaling approaches to web-scale data. The first things that people tried was scaling up the "good, old-fashioned" approaches that defined the rule-based systems (Chapter 4.4). Probably the most prominent system in this category is Start from Boris Katz at MIT. This system first launched in 1993. This, like many of the systems in this era, take an approach that's somewhat similar to to the old-fashioned AI approaches: parse the query with a and look it up in a knowledge base. For example, given the question:[1]

Who directed *Gone with the Wind*?

Becomes the lookup (Katz et al., 2002)

```
(get "imdb-movie" "Gone with the Wind (1939)"
                  "DIRECTOR") =>
    ("George Cukor" "Victor Fleming" "Sam Wood"))
```

Some of these can be looked up directly in datasets, but START further builds on the systems like LUNAR and BASEBALL we talked about in previous chapters. While those systems also need to turn the raw text of questions into a structured representation, START had to do that and deal with the messiness of Internet text. And its ability to *search* the Internet to find answers is clearly a huge influence on the subject of this chapter—*Watson*—and later Dense Passage Retrieval later (Chapter 9.3).

Transforming Questions to find the Answer

In the next chapters about neural language models, we will discuss how new models *transform* questions into numbers (vectors) to find or generate the answer. But before we get there, it is useful to talk about how we can transform questions into text that can better find the answer. We begin with the Question Answering using Statistical Models (Radev et al., 2001, QASM).

One way of thinking about question answering is that it's a translation problem: questions are asked in one language but the answers are found in another language. QASM's approach is to explicitly "undo" the process to find the correct response given an input clue.

In Chapter 10.1, we'll talk about the cheating scandal around the American game show *21*, where Charles van Doren got the answers in advance. Part of the mythology of *Jeopardy!* is that Merv Griffen said, why don't we just give them the answers! So you have clues like

[1] This is an impressive/complicated question because David O. Selznick removed the first director (George Cukor) to replace him with Victor Fleming (the credited director), but Sam Wood had to fill in when Fleming collapsed on set.

During WWII this computer scientist & code breaker converted his money into silver & buried it; he never found his buried treasure.

The correct response, which is phrased as a question, is "Who is Alan Turing?" In other words, the central conceit of Jeopardy! is that it converts questions into answers (and then the contestants need to give a response to that answer in the form of a question).

What QASM is doing is doing the reverse: it has a model that tries to turn questions into the search string that could find the answer to the question (we'll see modern versions of this when we talk about multihop question answering in Chapter 7). QASM can apply plenty of transformations: swapping words, deleting words, ignoring words (e.g., if you have Cleveland, Ohio and ignore Ohio, you'll find distracting things about US president Grover Cleveland), etc.

In the end, you have a model that creates specialized queries from the original question. Later on, when we talk about dense passage retrieval, those systems are doing something similar, except they're generating the query in a continuous dense vector space (Chapter 9.3). Doing it with real words like QASM is more interpretable: a human could understand what's going on, while a vector query is inscrutable it seems to work better.

But how does this allow *Watson* to answer questions? When a question comes in, *Watson* searches over previous *Jeopardy!* questions, Wikipedia, articles, and a select subset of the Internet to find evidence that can answer the question (Section 4.3).

Buzzing is important for Humans and Computers

Let's break this into two phases: **guessing** and **buzzing**. A guess is its best guess of what the answer could be: literally one of thousands of possible answers. A buzz is a binary decision of whether to trust whether this particular guess is correct or not.

Let's begin with an easy case: let's say you get a clue that's nearly identical to an existing clue:

DOUBLE D WORDS
Walk like a duck

Has appeared four times as a clue on *Jeopardy!*: 1987, 1996, 2001, and 2010. In addition to the above category, this clue also appeared under categories like **six-letter words** and **"WA"**. The guess process looks to see if it has ever seen this exactly clue before, it has, so it produces the guess waddle.

The buzzer needs to decide if it should trust this guess. It has a number that it associates with seeing the exact clue before; let's arbitrarily say that it's 3.0. It then compares that score with its threshold on when to buzz. Again, let's arbitrarily say that the threshold is 1.0. 3.0 is larger than 1.0, so it buzzes in with that guess (We'll talk more about how to set these numbers not-so-arbitrarily shortly).

But *Jeopardy!* doesn't have one kind of clue. There are many different types of question, and sometimes you get a clue from the category about what kind of question you will need to answer. Let's start with "potent potable rhyme time": from the name of the category, you know that both words in the response have to rhyme with each other and that it will have something to do with alchohol. So, to answer the clue:

Rice wine for the guy who rides a racehorse

you are essentially doing a constrained search: you can find individual pieces that fit the clues (e.g., rewriting synonyms to discover things like makkoli, brem, tapai, etc. for "rice wine" and "cowboy" and "knight" for people who ride a horse). So your guesser would generate lots of different possibilities, but from past experience your buzzer would give a very low score to anything that didn't have two words. For example, if the category is "rhyme time" but there's only one word in the answer, the score goes down by 5.0, which balances out even if the buzzer likes sake by itself as a response. This is in essence a constrained search: you search for individual components that fit with the clue until you get two things that match the clue and then rhyme with each other.

Using Features

Now how does the buzzer know to use this information? There are many little pieces that could go into thinking that a guess is good or not. These pieces of evidences are called features. While features aren't always active, when they are, they provide evidence of whether a guess is good. For example, you could have a feature (and Watson probably did) for when a category has a "quote" in it: this means that the text inside the quote should appear in the answer. E.g., if you have a category `"Ten"-letter words`, then the clue

Patrick Roy and Hope Solo played this position

would have many possible answers that might score highly if you didn't take the category into account: e.g., goalie. But the category has two pieces of information: that the answer should be ten letters long: goalie fails that but goalkeeper does. However, the additional quoted part of the category tells the buzzer that it should give a low score to any guess that doesn't have "ten" in it.

In a computer programming language, this could simply be an if statement: if ASCII character 34 is in this string, return 1. But just knowing that there's a quote in the category does't tell you which clues to favor. You would need to make the the feature more specific.

Let's go to a question that I got while I was on Jeopardy! And this shows that these aren't just a category-specific phenomenon…this also apply to individual clues as well. In a quite normal category about geography, this clue came up:

"G.I." hope you know that 0 degrees latitude & 180 degrees longitude is just east of this group, part of Kiribati

Notice that "G.I." is in quotation marks. This is a signal that the correct response will start with G.I., in this case the correct response is "Gilbert[2] Islands". So perhaps we need to have multiple features to capture whether our answer is consistent with this clue. To be clear, we don't know exactly what features Watson used, but it could look something like this:

- You might have one feature to indicate that there is a quotation mark in the clue.
- You could have another feature to show that the quote matches the candidate guess.
- Perhaps you have another feature that indicates whether the feature matches a multiword guess.

What made *Watson* such a *tour de force* is that humans had to come up with each of these features. This is not a one and done process. If you're building a system like this, you should look at things the system is still getting wrong and add a new feature to correct the issue. This is the same process that Watson used to build their statistical system. Maybe the system got confused when somebody has a quote from literature and you need to make sure that doesn't get counted or add a new feature to handle that case. It's very rare to get the features right the first time around.

This statistical approach has some advantages over the neural approach we'll see in the next chapter: it is easy to understand why a system is doing what it is. And it's easy to fix problems as they pop up.

Let's see how this might happen. For example, the constraint is that the correct answer starts with a particular letter. Often, this is straightforward in the category name: **Starts with "L"**. So you might have a feature that matches that exactly. But then you discover that this feature misses **Crossword Clues "P"**, and you can check that the feature didn't fire for that example and generalize that category.

The category that I feared the most was anagrams: where you need to rearrange the letters in the clue to get to the answer. This is actually easier for a computer than a human! If you have "anagram" in the category name, you create a feature that is true if the set of letters matches for both the clue and the guess.

Each of these can have different approaches that can generate guesses that might be the correct answer. And you might think that this is all encoded in the category. Not so! Sometimes the pattern reveals itself as you work your way through a category. For example **Let's Make a Movie Crossover!** requires you to combine two movie titles (or characters) together. So you would want a feature that would check for consistency with previous correct responses in the category.

[2] Named of course for Johnny Gilbert, "the voice of *Jeopardy!*". Actually not, it's far more confusing than that: it was named by a German admiral—Adam Johann von Krusenstern—who led the first Russian circumnavigation of the globe. Krusenstern recognized that the British Captain Thomas Gilbert had described each of the islands individually and applied the the name to the group of islands.

And for"normal" clues, generating a guess looks a lot like the reformulations that we saw in QASM:

In May 1898 Portugal celebrated the 400th anniversary of this explorer's arrival in India

A 400th anniversary of 1898 gets rewritten to May 1498, India gets transformed into Kappad, and then you can find something with the correct response: Vasco de Gama.

So the first phase of the Watson pipeline is to—in parallel—generate all sorts of guesses based on different interpretations of the question. Some try to solve anagrams, others look for rhymes, others run an IR query like the Start system we talked about before, others are doing transformations like the QASM approach we talked about before. And sometimes the approaches need to take a recursive approach, combining different subsystems to get to the right answer.

Deciding When to Trust a Guess: Which Features Matter?

From this you have dozens of possible guesses. How do you know which—if any—to select? Combining inputs to predict a value is often called a regression problem, and when the prediction is a binary decision, this is called a *logistic regression* problem. This can take into account how consistent the evidence matches the clue, how popular the response is, and it can even take into account things like Jeopardy!'s love of puns.

In the examples above, we assumed that we knew what the weights were for each of our features. For the moment, let's continue that assumption... but we'll try to improve them so that we get more buzzes right and fewer wrong. This will require a bit of math, so if you just want to jump ahead to the big picture,[3] go ahead to Section 6.7.

[3] This will be our first "technical" section. You don't need to read the technical sections to understand the rest of the non-technical parts of the book, but subsequent technical sections *will* depend on previous technical sections.

So what does it mean to get more buzzes right? First, we need to define a little terminology: we need to consider the correct examples $\mathcal{X}_c$ and all the wrong examples $\mathcal{X}_f$. We want all of the correct examples to have scores more than the threshold and all of the incorrect examples to have scores less than the threshold. Let's call the threshold b, and then we need to sum up all of the non-zero features of an example to compute the score of an example i:

$$\text{score}(\boldsymbol{x}_i) = \sum_j w_j x_{i,j}. \tag{6.1}$$

In other words, we take every feature, multiply it by the weight associated with it (e.g., $x_{i,j}$ might be does this clue have a category with a quote in it). We write this as a multiplication and not as just adding the feature weights directly because sometimes our features won't be binary. And this should look familliar: it looks a lot like the dot product that we first saw in the chapter on information retrieval (Chapter 4)), where we compute the *similarity* of two examples by multiplying the TF-IDF scores of two documents' words.

And this is a good segue to *why* we write this as a dot product rather than as a sum of all of the relevant feature weights. If all of the features of an example $x_{i,j}$ are binary, then you take the non-zero elements and add up the weights (since one times the weight just gives you the feature weights). But another very good feature might be how well the guess matches the clue (e.g., dot product between TF-IDF scores). So a way to generalize both the binary features and continuous features is to take the dot product of a vector representing all of the features and all of the weights to compute a score for the guess. The higher the score, the better the guess and the more likely the system will answer. And the weights will implicitly encode how important a feature is for making a decision.

But this doesn't answer the question of how we can learn what the weights should be. We need to look at the cases where the system has generated a lot of guesses and seen which of them are correct or not. From this, you can learn that when a letter is in quotes and it matches, that's a good sign for getting a guess right. More concretely we want the score (Equation 6.1) to be high for right guesses and low for wrong ones, and if it's not, we need to adjust the scores by changing the weights $\boldsymbol{w}$.

One missing piece of the explanation is how to turn our weights into probabilities. For that, we are going to use the logistic function. This is a function that can take any real number as an input and turn it into a probability between zero and one. When the input is zero, it provides a probability is 0.5, and when the input goes to positive infinity, the probability approaches one; likewise, when the input goes to negative infinity, the probability approaches zero. For convenience, we will call this function $\sigma(x)$, because some people call this the sigmoid function.

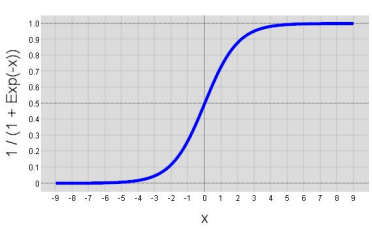

Figure 6.1: This function is called the sigmoid function because the way it transforms all inputs into values makes it look like an "S" shape, with 0 outputs to the left, increasing in the middle (an input of 0 right at the threshold gives an output of 0.5) and 1 outputs to the right.

For this, we are going to need an *objective function*: a mathematical expression that tells us how far we are from our goal. In this case our objective is to make the probability saying we should buzz—computed from our weights—for incorrect buzzes be as low as possible. So this means that we're going to transform our weights into a probability π_i between zero and one that gets

big when the weights pass the threshold b:

$$\pi_i = \sigma\left(\sum_j w_j x_{i,j} - b\right) \equiv p(\text{buzz} \mid \boldsymbol{x}) \equiv \frac{1}{1 + \exp\left(-\left(\sum_j w_j x_{i,j} + b\right)\right)}. \tag{6.2}$$

Since we only want to buzz when we're right, then we want π_i to be small (close to zero) whenever the buzz is wrong and big when the buzz is right (close to one). Alternatively, we could also say we want $1 - \pi_i$ to be big when the buzz is wrong, since this means that π_i is as close to zero. This alternate formulation is actually what we want since we will want to optimize a single function going in the same direction. Because we care about all examples, you would normally multiply the probability of joint events together, but this leads to a very small number (since probabilities are less than one), so for mathematical convenience we take the log:

$$\mathcal{L}' \equiv \underbrace{\sum_{i \in \mathcal{X}_c} \log \pi_i}_{\text{How right are we}} + \underbrace{\sum_{i \in \mathcal{X}_f} \log 1 - \pi_i}_{\text{How wrong are we}}. \tag{6.3}$$

So this overall expression (the log probability) tells us how good of a job our features are doing in telling us when to buzz.

Later in the book, objective functions like this will be called a *loss function*, so to be consistent with that, we will think about how to *minimize our mistakes*, so what we will do now (somewhat confusingly) is to minimize the negative of $\mathcal{L}'$, which we will call $\mathcal{L}$:

$$\mathcal{L} = -\sum_{i \in \mathcal{X}_c} \log \pi_i - \sum_{i \in \mathcal{X}_f} \log 1 - \pi_i. \tag{6.4}$$

Just to recap: Equation 6.2 is the probability of the model saying we should buzz on example i, Equation 6.3 combines all of those together to see how close we are to being right, and then Equation 6.4 flips it around so that the large it is, the wronger we are…we want to minimize that expression.

If you remember high school calculus, this is an optimization problem: you can compute the derivative (gradient, actually) of each of each of the variables you have control over—the feature weights $\boldsymbol{w}$—and then adjust those weights to decrease the loss function represented by $\mathcal{L}$ (Equation 6.4). When you did this in high school calculus, you could probably solve the equation for when the derivative was zero, set the variable to make the derivative zero, and then you're done. But that's not possible here, so we need to take little adjustments to $\boldsymbol{w}$ to push $\mathcal{L}$ down again and again.

An intuitive way of thinking about this is that the shape of $\mathcal{L}$ represents a hill. Where you stand on the hill is represented by how you've set $\boldsymbol{w}$: each dimension represents one dimension of this vector.[4] In two dimensions, one dimension is how far north–south you are and one dimension is how far east–west you are: and you're trying to get to the lowest valley you can. The

[4] The threshold b is also an additional dimension you need to optimize, but we'll gloss over this for now…you can always fudge it be imagining it an additional dimension of $\boldsymbol{w}$ that's always active for every example.

catch is that you can't really see what the whole landscape looks like; it's so foggy you can only see right around where you are. The intuition in this case is to walk "downhill", and this is exactly what the gradient of $\mathcal{L}$ gives you.

Now the problem is that computing $\mathcal{L}$ over your entire dataset is hard: you have to go over every single question in the *Jeopardy!* training set to compute its contribution to the gradient. The way that I like to think about it is that each example is a person you can ask "how do I go downhill". While you could ask lots of different people, if you ask literally everyone, that's probably going to be overkill. You can probably get a good sense of the direction by just asking a handful of example of which way you should go.

In the lingo, the sample of examples that you ask for the direction is the *minibatch*, and this overall process of asking a few (or even one) example for a direction is called *stochastic gradient descent* (Bottou, 2004).

Given that the math for the derivation is everywhere, I will skip it here (you can take a look at Chapter 5 of Speech and Language Processing by Jurafsky and Martin, which I often use in my classes and thus have adopted their notation), but it is so intuitive that if you look at it, it just "feels right". For a minibatch of size one, if you ask it which way you should go, it tells you to update[5] your weight w_i to be:

$$w_j^{(new)} = w_j^{(old)} + \lambda \frac{\delta \mathcal{L}}{w_j} = w_j^{(old)} - \lambda \underbrace{(\pi_i - y_i)}_{\text{error}} x_{i,j}. \tag{6.5}$$

[5] Because we're doing stochastic *descent*, we subtract the gradient from the current weight…if we were trying to make the objective function as large as possible, we'd add it to the current weights.

The most important part of the equation is the error: how far off your prediction for example j is. Remember that the true outcome y_i is binary: either you should have buzzed—$y_i = 1.0$—or you shouldn't have—$y_i = 0.0$. However, the prediction π_i is a probability and thus ranges between 0.0 and 1.0. The more right you are—the closer you are to y_i—the less the weights change. In other words, if you got this example exactly right, nothing changes at all. But the bigger your error is, the more you change. Exactly how much is controlled by the step size parameter λ, and the direction of the change depends on the sign of the error. If the error was positive, your prediction was too low, so you should increase all of the feature weights that caused you to say you should have buzzed when you didn't. If the error was negative, your prediction was too high, and so you slightly decrease all of the feature weights that caused you to get it wrong.

The other input to the update is how much of this feature the example has: $x_{i,j}$. To make sure your gradients don't get too big or too small, in practice we often standardize the feature weights so that they have mean zero and unit variance. However, because this chapter is trying explain how these things work (and we'll work through an example next), we won't do that to make the math more straightforward.

So let's see how this would work in a more real-world example. Let's say that you have a clue

US Cities

Its largest airport is named for a WWII hero.
Its second largest, for a WWII battle.

that Watson famously got wrong as a final *Jeopardy!* when it answered the famous American city Toronto. So in this case our buzzer input[6] is whether we should buzz on this clue with the response Toronto, for which we get a π_i. For the sake of concreteness, let's say π_i is 0.4…it's not high, but it *should* be zero because Midway and O'Hare airports are actually in the great city of Chicago. Now this is too high, so something went wrong with our features. Let's see what features were on and what their current weights in Table 6.1.

[6] The calculus for Final *Jeopardy!* is different: you always provide a guess, so there's no buzzing. However, you still want the best answer to have the highest probability, so we can still work through this example. Given the debugging output *Watson* wasn't confident, so it probably wouldn't have "buzzed in" during normal play.

Table 6.1: Imagined features used in logistic regression for a wrong example.

Feature	Weight w_i	x_i
IR Score	2	0.1
Toronto compatability with Category	1	0.2
Knowledge Base (# Airports in Toronto)	0.1	2
Previous responses (Toronto)	0.005	40
Bias	-0.8	

These are reasonable features: the IR score encodes how similar this is to questions that you've seen before; the category feature encodes whether the answer is compatable with the category "US cities" (and while you imagine that it isn't great, you could imagine substituting "American" for "US", which would make Toronto a lot more plausible); the knowledge base feature checks to see how many airports are in Toronto; and the final feature checks to see if Toronto has been an answer before (you might imagine that more frequent answers in the past will appear again).

There are lots of other features that we're not seeing because they're zero, e.g., the feature that checks to see if there's a quotation mark in the category doesn't see one, so it's zero and we can safely ignore it along with every other feature that didn't fire.[7] And this makes sense: the features that aren't on for this feature had nothing to do with the mistake, so they shouldn't be punished or rewarded.

[7] Lest there be any doubt, I am making up these weights and features for a simple example… *Watson* had many more and better features. Most unbelievably, the first four features are non-zero for this example and the rest are zero. Finally, you normally don't have feature weights with such round numbers, but it makes the example cleaner.

First, lets see how these features translate into a probability:

$$\pi_i = \sigma\left(\underbrace{2\cdot 0.1}_{\text{IR}} + \underbrace{1\cdot 0.2}_{\text{Category}} + \underbrace{0.1\cdot 2}_{\text{KB}} + \underbrace{0.005\cdot 40}_{\text{Prev}} - \underbrace{0.8}_{\text{bias}}\right) \tag{6.6}$$

$$= \sigma\left(0.2 + 0.2 + 0.2 + 0.2 - 0.8\right) = \sigma\left(0.0\right) = 0.5 \tag{6.7}$$

From Equation 6.5, the error is $0.5 - \pi_i$ should have been zero, but it was 0.5—and if we assume that our step size[8] is 1.0, we can now write the overall update

[8] In practice, this would be a really bad step size—it's way too large, you're likely to jump over the valley in the objective function you're looking for—but we'll go with this because it makes the math easier.

$$
\begin{aligned}
\boldsymbol{w}^{(new)} &= \boldsymbol{w}^{(old)} - \lambda \nabla_{\boldsymbol{w},b} \mathcal{L} \qquad (6.8)\\
&= \begin{bmatrix} 2 \\ 1 \\ 0.1 \\ 0.005 \\ -0.8 \end{bmatrix} - \lambda \begin{bmatrix} \frac{\partial \mathcal{L}}{\partial w_1} \\ \frac{\partial \mathcal{L}}{\partial w_2} \\ \frac{\partial \mathcal{L}}{\partial w_3} \\ \frac{\partial \mathcal{L}}{\partial w_4} \\ \frac{\partial \mathcal{L}}{\partial b} \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0.1 \\ 0.005 \\ -0.8 \end{bmatrix} - 1.0 \begin{bmatrix} 0.5 \cdot 0.1 \\ 0.5 \cdot 0.2 \\ 0.5 \cdot 2 \\ 0.5 \cdot 40 \\ 0.5 \cdot 1 \end{bmatrix} = \begin{bmatrix} 1.95 \\ 0.75 \\ -0.9 \\ -19.995 \\ -1.3 \end{bmatrix}.
\end{aligned}
$$

This shows the dangers of having unbalanced feature weights and too large step sizes. This change is overall too big; you want to be taking little steps on your objective function, not taking huge leaps—you might jump over the "right answer" in front of you and not fix the real problem in this example: that Toronto is inconsistent with the category "US cities".

Feature Engineering

However, these updates can only update the features that you have. They cannot add new features nor can they fix problems intrinsic in the features. When you're starting out building a system like this, you usually don't have very many features. But as you go through the process of testing your initial system, you'll see lots things that your system gets wrong but *shouldn't*: your system has all of the evidence it needs to do what it needs, but cannot actually get there.

Let's focus on the Toronto question a little bit more. We already talked about how the feature itself might need some debugging (e.g., making sure that "US" doesn't expand to cover all of the American continent), but how could we add additional features that could handle this kind of phenomena better?

In a 2010 Tournament of Champions game, there was a category with the first clue:

CELEBRATIONS OF THE MONTH
D-Day anniversary and Magna Carta day

When the category was revealed, the host Alex Trebek said "you have to name the month", but *Watson* didn't get that hint. In a presentation from IBM, they showed that Watson got that clue wrong (Figure 6.2).

Clue	Type	Watson's Answer	Correct Answer
D-DAY ANNIVERSARY & MAGNA CARTA DAY	day	Runnymede	June
NATIONAL PHILANTHROPY DAY & ALL SOULS' DAY	day	Day of the Dead	November
NATIONAL TEACHER DAY & KENTUCKY DERBY DAY	Day/month(.2)	Churchill Downs	May
ADMINISTRATIVE PROFESSIONALS DAY & NATIONAL CPAS GOOF-OFF DAY	day / month(.6)	April	April
NATIONAL MAGIC DAY & NEVADA ADMISSION DAY	day / month(.8)	October	October

Figure 6.2: Slide from IBM *Watson* on how a feature to predict the answer type as it works it way through a category's column can improve its ability to buzz in correctly. (Figure Credit: IBM)

Something that is particularly unique to *Jeopardy!* and not to the other QA settings we've looked at is that the category (they text that appears atop the column a clue appears in) is a huge constraint on what the correct responses are. A large part of the Jeopardy! buzzer is figuring out the "lexical answer type": what kind of thing can the answer be and only buzzing in on the things that are consistent with that type.[9] Part of this is is looking at clues in the clue. This chapter has used a lot of example clues like "this woman" (Sue Grafton), "this code breaker" (Turing), "this position" (goaltender), or "this group" (Gilbert Islands). These are hints about what kind of response is being sought. And lest you think this is an artifice of the skilled writers of *Jeopardy!*, this also appears in more "natural" questions like those people ask Google (Chapter 8.2).

[9] The Watson project called these signifiers lexical answer types, but the Quizbowl community (Section 5.2) uses the term "pronoun" or "identifier". I prefer lexical answer type, however, so I'll stick with that.

And part of what made Watson a good *Jeopardy!* Player is that it would learn as it explored the category. After getting a couple of months wrong, Watson can learn that all of the answers should be a month. While we don't know for sure how this was implemented in detail, we can imagine that there is a feature that suggests whether the clue is consistent with an answer type of "day" or "month" (e.g., this clue is consistent with a "month" as an answer, and June is a month). Resources like WordNet (?) can help with this: it provides a tree-like organization for all of the nouns in English so that you can check whether "June" is a month, whether a "schnauzer" is a dog, or whether "pride" is an emotion. And information from the current column could also be included in this, either directly in the computation of the feature or directly encoding something like "given the guess April, how many of the previous responses in the category are consistent with that type".

One thing that made *Watson* particularly groundbreaking was that it not just computed raw accuracy but also compared against human performance. This chart showed how Watson progressed in different iterations of the system, inching up to the cloud of *Jeopardy!* champions. Ken Jennings is in red there, still clearly dominating even the final version of Watson. This comparison is even more important today (Chapter 10) as people claim that AIs have super-human, but the lessons of *Watson* can help inform how we we can judge whether these judgements are fair and reasonable. So was the *Watson* match "the real deal"?

6.8 *This Game is Rigged, I Tell Ya!*

The nominal success of *Watson* has been well documented (not least by IBM itself, who rightly celebrated the great technical achievements and the great show they put on); however, things were not perfect…the game was rigged. It's useful to go over the lifecycle of an entire question: how it was written, how it's communicated, how players answer, and how the game unfolds afterward. At every stage, there's a slight benefit to the computer, which taken together makes this an unfair competition.

This is a problem! First, it's a problem scientifically because we want to have fair comparisons of human vs. computer intelligence. More importantly, I want to have my turn having my question answering robots sit opposite against trivia whizzes (Chapter 12), and I can't do that if everybody thinks that Watson's spin on *Jeopardy!* settled the issue (and it hasn't).

But first, in case you don't know how *Jeopardy!* works, we'll review that. However, if you've calculated a Coryat score before, you can go ahead and skip ahead to Section 6.8.

The Pool of Questions

Part of the agreement between *Jeopardy!* and IBM was that the competition would take place on normal, written questions. In the media coverage of the competition, this focused on avoiding video and picture daily doubles (fairly reasonable, but we'll discuss how multimedia questions might be more fun in Chapter 7.5). However, this causes two problems: the questions are too easy and do not necessarily challenge computers.

So what makes up "normal" questions? Every game of *Jeopardy!* has questions that range in difficulty. Because it's a television show, many questions are easy enough that the average viewer at home can get them. Moreover, the humans on the stage with Watson are not normal contestants. Ken Jennings is certifiably the greatest of all time (Low, 2020, GOAT) *Jeopardy!* player, and Brad Rutter isn't bad himself.

The average "normal" *Jeopardy!* contestant, including not so great players like yours truly, know a large majority of the clues. For top players like Brad and Ken, they know—with a handful of exceptions–*all* the clues. In a one-on-one fight with normal clues, Ken and Brad would be fighting over every clue: it would come down to who could buzz first.

This isn't fun to play. Nor is it fun to watch. This is why *Jeopardy!*'s tournament of champions is played on much more difficult[10] clues (Harris, 2006). Nonetheless, this is the battlefield where Watson won: "normal" questions that didn't challenge the human players. Instead, it all came down to the buzzer.

[10] The difficulty ratchet isn't one way; all "special" matches use designated questions: "Celebrity *Jeopardy!*" are easier (as mocked by categories like "States that end with Hampshire" or clues like "You wear these on your face to help you see better" on *Saturday Night Live*), and "College *Jeopardy!*" isn't necessarily easier but does have more college football and popular music. Questions are tuned to expected players' abilities. IBM did not want the computer to be targeted, lest the questions be adversarial against the computer. Chapter 10.6 discusses what this would look like if we specifically *wanted* that to happen.

John Henry vs. the Buzzing Machine

Unlike QB (Chapter 5.2), while *Jeopardy!* also uses signaling devices, these only work *once the question has been read in its entirety*; Ken Jennings (also a former QB player while he was a student at BYU) himself explains it on a *Planet Money* interview (Malone, 2019):

> **Jennings:** The buzzer is not live until Alex finishes reading the question. And if you buzz in before your buzzer goes live, *you actually lock yourself out for a fraction of a second.* So the big mistake on the show is people who are all adrenalized and are buzzing too quickly, too eagerly.
> **Malone:** OK. To some degree, *Jeopardy!* is kind of a video game, and a *crappy video game where it's, like, light goes on, press button*—that's it.

Jennings: (Laughter) Yeah.

Jeopardy!'s buzzers are a gimmick to ensure good television; however, QB buzzers discriminate knowledge.

So how does this interact with Watson? Watson receives all of the clues electronically and likewise gets an electronic signal to know when it is safe to buzz in, then computes a probability of being right and buzzes in if it's above that threshold (Section 6.7). In contrast, humans have to either guess when it is safe to buzz or wait for a light to turn on.

Jeopardy! gurus explicitly advise new players not to wait for the light—your puny human reflexes are too slow. Indeed, one of Ken Jenning's strengths was his uncanny ability to internalize the cadence of Alex's voice and when a tech would activate the buzzer (Jennings, 2006). In contrast, Watson is literally an electromechanical buzzing machine that could get first crack at every question it wants.[11]

[11] In practice, this is not always true. Because Watson computed its responses in real time, it could not come up with a response in time for particularly short questions.

Unfortunately, despite subjecting everyone within earshot to these rants, the computer science community thinks that the question is settled: computers are better than humans at answering questions. This is despite Ken basically saying that it did, indeed, come down to the buzzer.

Moreover, what makes for a "normal" difficulty question for a human does not always apply to a computer. Let's first talk about what makes a clue easier for a computer and then we'll talk about what makes a clue harder for a computer.

Easy for a Computer. When I appeared on *Jeopardy!*, my final *Jeopardy!* was:

> After this woman's death, her daughter wrote, "As far as we in the family are concerned, the alphabet now ends at Y"

All of us got the question right; it just so happened that *Jeopardy!* used a very similar clue that aired as we were recording:

> "G" is for grand master as well as this woman who received the 2009 Grand Master Award.

The correct response is of course Sue Grafton. For poorly read contestant like myself, only studying previous clues allowed us to get the answer right. I've never read a Grafton book, but I know she writes mystery books and has titles of the form *"A" is for Alibi* (and that's the only title I can think without looking at Wikipedia).

For Watson, this is "memorization" is trivial: a single letter implies that the answer is Grafton. But just like you should not think that I'm smart for getting lucky to have seen a reused clue, you should not praise Watson for finding near-repeat questions. And it's not just trivia games: Google's dataset of questions (which we'll talk about in more Chapter 8.2) have many identical

questions (Lewis et al., 2021), which makes it an imperfect yardstick. Moreover, Watson can also store the entirety of Wikipedia, easily looking up capitals, authors, etc.

Indeed, when a computer can find *an exact quote* (as was in my final *Jeopardy!* clue), the question becomes even easier. Then the computer just needs to find the appropriate article that contains the quote and then just find whatever entity is mostly likely to be a *Jeopardy!* answer.

Where systems like Watson struggle are on computation, matching novels and movies to plots, combining multiple clues, lateral thinking, and wordplay (Kaushik and Lipton, 2018). And this is not just a matter of degree: computers struggle with *all* such questions, even if they're in the top row of *Jeopardy!* While a computer is theoretically good at math, the kinds of programs that answer trivia questions struggle answering match questions with numbers in the double digits (Wallace et al., 2019b).

This is why the goal of AI is *general* artificial intelligence (Chapter 3): while we can build specialized systems for *Jeopardy!* clues or math problems, unlike a reasonably smart human, a single program can't "do it all". Unlike for human contestants, the "difficulty" of *Jeopardy!* questions for a computer has no relationship to the nominal value. We talk about how Facebook/Meta dealt with this problem in Chapter 10.6: tell the authors what's hard for a computer.

6.9 Two Nice Guys, One Computer with no Shame

Given the "too easy" questions and the buzzer, what does this actually mean for gameplay? A question comes in, and Watson has the choice of answering it or not: it can win every race to the buzzer if it wants. Then, of the things it cannot answer, Ken and Brad fight over the scraps. Thus, for a computer to win this competition, it needs only to be able to answer a third of the questions correctly.

Now, the *Jeopardy!* nerds reading the book (I love you all), will point out that this isn't true, because the clues are not weighted equally: some are worth more than others. However, as we discussed above, what's difficult for a computer isn't always difficult for a human (and *vice versa*), so it really is a random third of the questions. While a good human player might be weak on the buzzer and be confident that if they know more they'll win the harder clues, this isn't true for a human facing off against *Watson*.

Moreover, the computer has no shame: it uses a strategy called the "Forrest Bounce" (Rogak, 2020, more infamously associated with James Holzhauer and Arthur Chu). Rather than going through the categories top to bottom (easy to hard), Watson goes through the clues somewhat randomly, searching for Daily Doubles and trying to optimize its score. Again, there's nothing wrong with this—it's the optimal strategy! But if it's the "right" way to play the game, why doesn't every human do it?

That's because humans want to follow social norms. The producers of the

show tell you up and down that you shouldn't play the game like that, and you don't want to make them unhappy with you… they can make your life miserable. I remember watching *Jeopardy!* with my grandmother and when someone hunted for the Daily Double, she would always say "who does he think he is" (and it was always a he). Alex Trebek also wasn't a fan (Marchese, 2018):[12]

> When the show's writers construct categories they do it so that there's a flow in terms of difficulty, and if you jump to the bottom of the category you may get a clue that would be easier to understand if you'd begun at the top of the category and saw how the clues worked. I like there to be order on the show, but as the impartial host I accept disorder.

And nobody, nobody, wants to make Alex unhappy.

Ken would sometimes do a little hunding for the Daily Double against a particularly formiddable opponent, but he would normally be well-behavied so as not to upset the powers that be. *Watson*, however, was a soulless machine; and having a machine on the stage was no exciting that nobody faulted it for its strategy. If anyone is to blame, it's probably Gary Tesauro; adding his strategy for playing the board increased Watson's win percentage considerably (Tesauro et al., 2013). But if you put him in a room with the withering stares of *Jeopardy!* producers (or worse, Alex Trebek) and that code would be deleted in no time.

6.10 *The Legacy of Watson*

Let's review Watson's appearance on *Jeopardy!*:

1. all questions are of "normal" difficulty;
2. thus the two human contestants know nearly all of the clues; but
3. Watson can win the buzzer race whenever it wants.

If Watson wins such a match, does that mean that it is superior to these supurb humans?

I hope that you are reluctant to answer "yes" (not just because *Jeopardy!* has trained you to respond to answers with questions). Perhaps I've planted a seed of doubt: *no, we cannot yet conclude computer superiority* from this experiment. And this is not the end of the story for judging whether computers are superhuman in their intelligence. Since then, we've seen claims that computers are smoking lawyers in taking the LSAT (**?**) or is better than radiologists in reading X-rays (**?**). *Watson* was just the beginning of the story, as since then we've seen both a tremendous improvement in the technologies that drive AI (we review these advances in Chapter 9.4) and a greater interest in measuring how smart these models are. *Watson* was just the beginning of the story, in both respects.

In the next chapters, we go beyond the methods that *Watson* used to the modern QA has *actually* reached possible parity with humans and how to

[12] Rogak (2020) quotes a saltier take Trebek offered to Howard Stern: "It only works, dickweed, if you know the correct response to everything that's up there."

actually measure how far AI has come since Ken Jennings lost to *Watson* (Chapter 10).

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